How do you manage aggression in severe autism?

Evidence searched on 2026-10-17 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line Functional neurosurgery is under consideration for refractory cases after conventional treatments fail, but its role remains highly specialized and controversial.

Most important recent finding Functional neurosurgery may offer clinically meaningful symptom relief for carefully selected patients with severe autism who have exhausted conventional treatments, though ethical and regulatory oversight remains critical. [1]

Immediate context Pharmacological and psychosocial interventions (e.g., CBT, physical activity) are the established first-line approaches for aggression in autism, supported by modifiable protective factors. Neurosurgical options are experimental and reserved for extreme cases where no other interventions work. No direct evidence supports virtual reality or sandplay therapy for aggression specifically.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)."	DailyMed

Latest findings

- **Intensive outpatient behavioral treatment for children with severe externalizing behavior and intellectual disability (2026 Jul-Aug 01)**

A retrospective electronic health record review of **80 children with autism and co-occurring intellectual disability** found a **90.74% reduction in externalizing behavior** (including aggression in 87.5% of participants) at discharge. The primary limitation was the lack of a control group, making causality uncertain [2].

- **High-dose propranolol for severe aggression in autism (2024 Sep-Oct 01)**

A **pilot, double-blind, placebo-controlled crossover study** in **6 children and young adults with severe autism and chronic aggression** (ages 12–19) reported a **50% reduction in aggression symptoms** (CGI-I) and a **37% reduction** (ABC-I) during propranolol treatment. The small sample size and lack of long-term follow-up limit generalizability [3].

- **Deep brain stimulation (DBS) for refractory aggression in pediatric autism (2024 Jul 30)**

A **meta-analytic review** of **65 pediatric patients with autism and severe intellectual disability** found **94.2% clinical improvement in aggression** after DBS, with the **posteromedial hypothalamic nuclei (pHypN)** as the safest target. However, **13/65 patients experienced adverse effects**, and most studies lacked >=12 months of follow-up [4].

- **Clozapine for severe behavioral dysregulation in autism (2026)**

A **case report** of an **18-year-old adolescent with autism, schizoaffective disorder, and moderate intellectual disability** showed **decreased aggression and self-injury** after **14 months of clozapine treatment**, with no

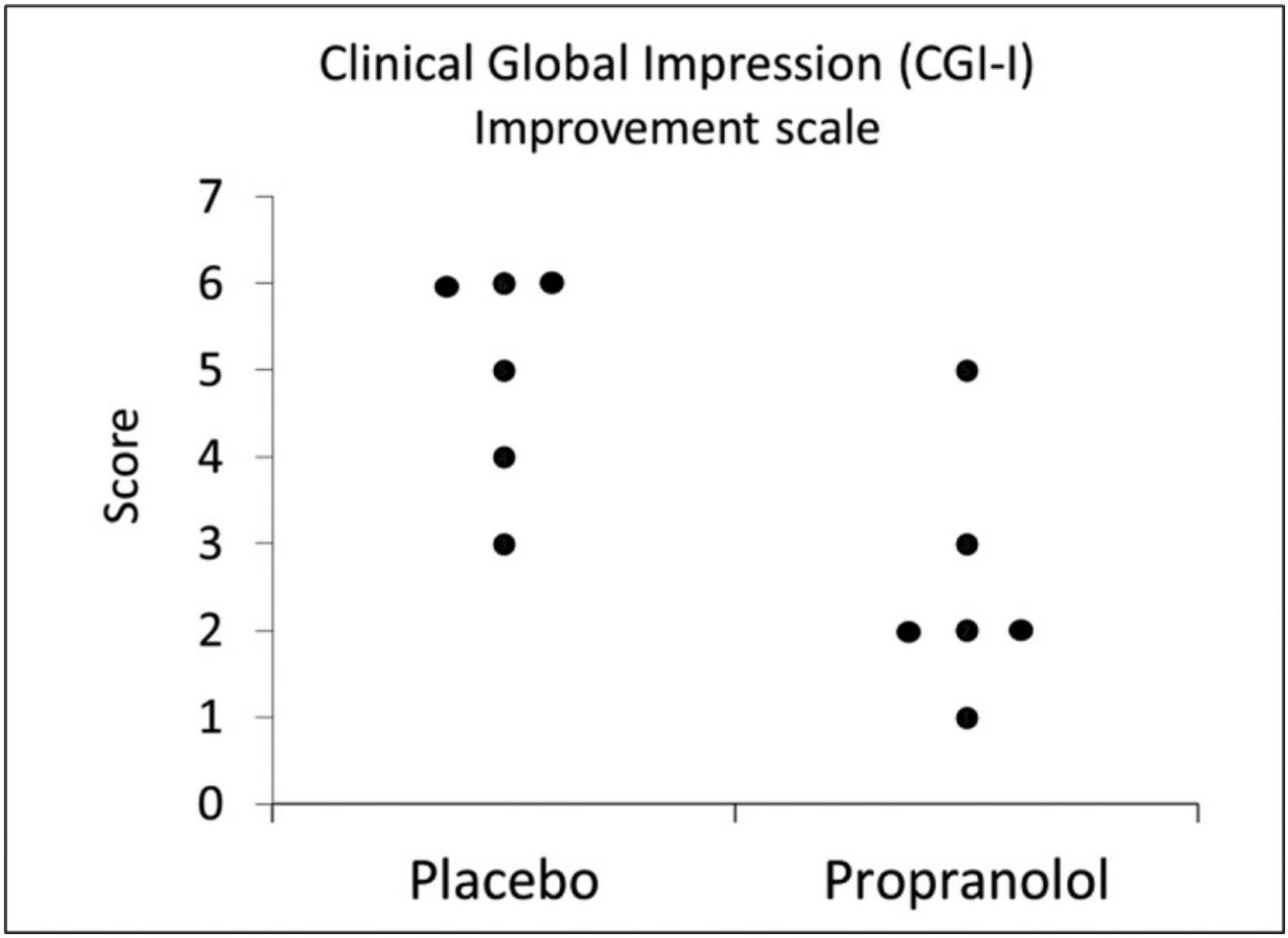
significant side effects. The lack of a control group and single-patient design preclude broader conclusions [5].

- **Atypical antipsychotics for irritability in autism (2025 May 21)**

A **network meta-analysis** of **13 studies (906 participants)** found **risperidone and aripiprazole reduced irritability** (mean difference: **-7.89 and -6.26**, respectively) compared to placebo in the short term, but **effects on aggression were uncertain**. Low-to-moderate certainty limited confidence in these findings [6].

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Intensive behavioral treatment	Evidence-supported but limited	In children with autism and intellectual disability, an electronic health record review showed a 90.74% reduction in externalizing behaviors (including aggression) at discharge, with 82.5% meeting caregiver-directed goals	[2]
Functional analysis (FA) for severe aggression	Evidence-supported but limited	FA is considered essential for treating severe challenging behaviors (e.g., aggression, self-injury) in autism, with guidelines supporting its use in clinical decision-making	[9]
High-dose propranolol for severe aggression	Emerging	A pilot double-blind crossover study in adolescents/young adults with severe ASD-related aggression reported a 50% reduction in symptoms (CGI-I) and a 37% reduction (ABC-I) with propranolol vs. placebo	[3]
Deep brain stimulation (DBS) for refractory aggression	Experimental	A meta-analytic review of pediatric DBS for intractable aggression in ASD with intellectual disability found a 94.2% clinical improvement rate in 48/51 patients with >=12-month follow-up, though adverse effects were reported in 13/65 subjects	[4]
Clozapine for severe behavioral dysregulation	Experimental	A case report of an adolescent with comorbid ASD, schizoaffective disorder, and intellectual disability showed decreased aggression and improved social skills with clozapine over 14 months, with no significant hematologic adverse events	[5]
Atypical antipsychotics (risperidone/aripiprazole) for irritability	Evidence-supported but limited	Network meta-analysis showed risperidone and aripiprazole may reduce irritability vs. placebo in children with ASD (low-certainty evidence), but uncertainty remains for aggression	[6]



What this figure shows: This figure compares individual Clinical Global Impression Improvement (CGI-I) scores for participants receiving placebo versus high-dose propranolol, showing lower (better) scores in the propranolol phase. It demonstrates a trend toward symptom reduction with propranolol, which bears on its potential efficacy for managing aggression in severe autism.

How it helps with treatment options: This figure helps doctors compare the relative symptom improvement between propranolol and placebo, supporting its consideration as a treatment option for aggression in severe autism.

How to read this figure: The y-axis shows CGI-I scores (0 = very much improved, 7 = very much worse), with individual data points for each participant in the Placebo and Propranolol groups. Lower scores indicate greater improvement in aggression symptoms. The figure compares individual responses between the two phases.

Why this figure was chosen: The model's reason, from the caption: “Endpoint CGI-I per study phase”. Checks passed: the paper was judged directly relevant to the question; the article states an open licence; the model's five-part relevance test was passed (helps answer, clinically meaningful, adds to the report, not decorative, shows the kind of information the question asks for); the picture is large enough to read on the page; a vision model confirmed that the image matches its caption and is readable.

FIGURE 1. FIGURE 1. Endpoint CGI-I per study phase.

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Used in practice: a retrieved review or page describes the drug for this condition, with no FDA label or guideline for it (see the category in Key evidence). Doses appear only where an FDA label or source gives them for the use shown; each dose entry is a short summary of the label's dosing section for this condition, with every number checked against the label text (full text in the DailyMed link); individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Key evidence
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)." [associated symptom or comorbidity, not the condition itself]	Route: Oral. For patients weighing less than 20 kg, start with a dose of 0.25 mg per day. After at least four days, increase the dose to 0.5 mg per day, maintain it for no less than 14 days, and then raise it in steps of 0.25 mg per day every two weeks or more if needed. For patients weighing 20 kg or more, begin with 0.5 mg per day, then increase to 1.0 mg per day after at least four days and keep it for no less than 14 days, raising it in steps of 0.5 mg per day every two weeks or more if required. The usual dose range is 0.5 mg to 3 mg per day, and this applies to children and adolescents weighing at least 15 kg.	FDA label: Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years). DailyMed label FDA-approved for the use shown; label effective 2026-06-25.
clozapine	as stated in the quoted source sentence	Not stated in the retrieved source for this use	"In this case report, we present the first 14 months of an 18-year-old adolescent male with comorbid diagnoses who exhibits extremely aggressive and self-injurious behavior and is treated with a combination of clozapine and behavioral interventions." [5] (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)
aripiprazole	as stated in the quoted source sentence	"May 14, 2025 · The starting dose of aripiprazole (brand name Abilify) for children and adolescents with autism spectrum disorder (ASD) is generally 2 mg taken once daily." source (from a retrieved page, not an FDA label)	"The patient improved gradually and was discharged after four days in intensive care and 14 days in the ward, on aripiprazole and clonidine." [7] (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Dose / route (if sourced)	Key evidence
propranolol	decreasing aggression in individuals with ASD	Not stated in the retrieved source for this use	"CONCLUSION: These results suggest that propranolol is an effective option in decreasing aggression in individuals with ASD." [3]
methylphenidate	ADHD-like symptoms [associated symptom or comorbidity, not the condition itself]	Not stated in the retrieved source for this use	"Autism Spectrum Disorders (ASD) are a highly heterogeneous neurodevelopmental conditions defined by ... to manage comorbid irritability and aggression, stimulants and non-stimulants (e.g., methylphenidate, atomoxetine, clonidine, guanfacine) to treat ADHD-like symptoms, and melatonin for sleep disturbances." [8]
atomoxetine	ADHD-like symptoms [associated symptom or comorbidity, not the condition itself]	"Oct 1, 2026 · The dose is increased after a minimum of 3 days to a total daily dose of 80 mg as a single dose in the morning or divided in 2 doses (morning and late afternoon)." source (from a retrieved page, not an FDA label)	"Autism Spectrum Disorders (ASD) are a highly heterogeneous neurodevelopmental conditions defined by ... to manage comorbid irritability and aggression, stimulants and non-stimulants (e.g., methylphenidate, atomoxetine, clonidine, guanfacine) to treat ADHD-like symptoms, and melatonin for sleep disturbances." [8]
clonidine	irritability and aggression [associated symptom or comorbidity, not the condition itself]	"Initial dose: 0.1 mg orally 2 times a day (morning and bedtime) 2." source (from a retrieved page, not an FDA label)	"Autism Spectrum Disorders (ASD) are a highly heterogeneous neurodevelopmental conditions defined by ... to manage comorbid irritability and aggression, stimulants and non-stimulants (e.g., methylphenidate, atomoxetine, clonidine, guanfacine) to treat ADHD-like symptoms, and melatonin for sleep disturbances." [8]
guanfacine	ADHD-like symptoms [associated symptom or comorbidity, not the condition itself]	Not stated in the retrieved source for this use	"Autism Spectrum Disorders (ASD) are a highly heterogeneous neurodevelopmental conditions defined by ... to manage comorbid irritability and aggression, stimulants and non-stimulants (e.g., methylphenidate, atomoxetine, clonidine, guanfacine) to treat ADHD-like symptoms, and melatonin for sleep disturbances." [8]

Recently approved or updated (FDA, EMA and CDSCO announcements)

No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
methylphenidate	"ABUSE, MISUSE, AND ADDICTION Methylphenidate hydrochloride extended-release has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction."
atomoxetine	"SUICIDAL IDEATION IN CHILDREN AND ADOLESCENTS Atomoxetine increased the risk of suicidal ideation in short-term studies in children or adolescents with Attention-Deficit/Hyperactivity Disorder (ADHD). Anyone considering the use of atomoxetine in a child or adolescent must balance this risk with the clinical need. Co-morbidities occurring with ADHD may be associated with an increase in the risk of suicidal ideation and/or behavior."
clozapine	"SEVERE NEUTROPENIA; ORTHOSTATIC HYPOTENSION, BRADYCARDIA, AND SYNCOPE; SEIZURE; MYOCARDITIS, PERICARDITIS, AND CARDIOMYOPATHY; INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Severe Neutropenia Clozapine Orally Disintegrating Tablets (Clozapine ODT) have caused severe neutropenia which is associated with an increased risk of serious and fatal infections. Prior to initiating Clozapine ODT treatment, obtain baseline ANC(s)."
aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
Risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Risperidone tablets are not approved for the treatment of patients with dementia-related psychosis."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: propranolol, clonidine, guanfacine.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07433517	VGN-Ex05e	Phase 1	Not yet recruiting (2026-02-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04047355	Propranolol	Phase 2	Completed (2025-06-04)	Completed; results posted (outcome not retrieved here).
NCT05056415	Everyday Life Rehabilitation (ELR); Treatment as usual (TAU)	Phase not applicable	Completed (2024-09-19)	Completed; results not posted/retrieved.
NCT00889538	placebo; glutathione	Phase not applicable	Completed (2016-12-07)	Completed; results not posted/retrieved.

6 further registry record(s) were retrieved and are not shown (judged not to be treatment trials for this question).

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Meza et al. (2025)**, meta-analysis: “Lurasidone probably results in little to no difference in irritability compared to placebo in the short term (MD -1.30, 95% CI -5.46 to 2.86; 13 studies, 906 participants; moderate-certainty evidence).” [6]
2. **Herrera-Pino et al. (2024)**, meta-analysis: “Thus, in 48/51 a clinical improvement of patients was estimated (94.2%), with a considerable overall effect size (OAS: d = 4.32; MOAS: d = 1.46).” [4]
3. **London et al. (2024)**, randomized controlled trial: “The CGI-I indicated a 50% reduction in symptoms in the propranolol phase, while the ABC-I indicated a 37% reduction in comparison to placebo.” [3]
4. **Moleón-Ruiz et al. (2026)**, narrative review: “When applied within specialized centers and under rigorous clinical, ethical, and regulatory oversight, functional psychosurgery may offer clinically meaningful symptom relief and improved quality of life for carefully selected patients who have exhausted conventional treatments.” [1]
5. **Ugwuocha et al. (2026)**, narrative review: “The framework offers a practical basis for guiding restrictive-practice reduction, quality improvement, equity-sensitive monitoring and future empirical research in inpatient mental health care.” [10]

Conflicting evidence

The **London et al. (2024)** pilot study reported a significant reduction in aggression with high-dose propranolol (50% reduction on CGI-I) in a small sample of six children with severe ASD, while **Trauner et al. (2025)** found that cannabidiol (CBD) showed clinically evident improvement in only 2/3 of participants, with a strong placebo effect complicating interpretation. No direct comparison between propranolol and CBD for aggression in ASD was retrieved, so the reason for the discrepancy is not established.

What remains under investigation

- **Long-term outcomes of deep brain stimulation (DBS)**: While **Herrera-Pino et al. (2024)** reported short-term improvements in aggression with DBS, only 51/65 patients had >=12 months of follow-up, leaving long-term efficacy and adverse effects uncertain.

Retrieval was incomplete (web_guideline_op_12, web_op_15); the absence of drugs or approvals cannot be concluded.

- **Dose-response and safety of clozapine**: The **Ralph et al. (2026)** case report highlighted clozapine’s potential for severe behavioral dysregulation in ASD, but no controlled trials or large-scale studies were retrieved to confirm efficacy or safety profiles.
- **Transdiagnostic risk factors for aggression**: **Estes et al. (2026)** identified sleep difficulties, speech/language delays, and gastrointestinal issues as associated factors, but no intervention studies were retrieved to address these modifiable risks.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages; for 5 open-access papers the methods, results and conclusion were also read, and study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- 15 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.
- The guideline-organisation search did not respond fully, so it may be incomplete; the absence of evidence cannot be concluded from it.

- The web search did not respond fully, so it may be incomplete; the absence of evidence cannot be concluded from it.

Sources

1. Moleón-Ruiz Á, Álvarez de Toledo P, Martín-Bejarano M, et al. Functional psychosurgery in psychiatry: indications and preoperative evaluation. A narrative review. *Frontiers in psychiatry*. 2026-08-19. PMID 42689152. DOI 10.3389/fpsyt.2026.1844401.
2. McMahon MXH, Scheithauer MC, Herrod JL, et al. Intensive Outpatient Treatment for Patients With Severe Externalizing Behavior and Intellectual and Developmental Disability: An Electronic Health Record Review. *Journal of developmental and behavioral pediatrics : JDBP*. 2026 Jul-Aug 01. PMID 42377401. DOI 10.1097/DBP.0000000000001498.
3. London EB, Zimmerman-Bier BL, Yoo JH, et al. High-Dose Propranolol for Severe and Chronic Aggression in Autism Spectrum Disorder: A Pilot, Double-Blind, Placebo-Controlled, Randomized Crossover Study. *Journal of clinical psychopharmacology*. 2024 Sep-Oct 01. PMID 39174017. DOI 10.1097/JCP.0000000000001895.
4. Herrera-Pino J, Benedetti-Isaac J, Ripoll-Córdoba D, et al. Effectiveness of deep brain stimulation on refractory aggression in pediatric patients with autism and severe intellectual disability: meta-analytic review. *BMC pediatrics*. 2024 Jul 30. PMID 39080575. DOI 10.1186/s12887-024-04920-x.
5. Ralph P, See S, Faust A, et al. Clozapine use in an adolescent with comorbid diagnoses of autism spectrum disorder, schizoaffective disorder, moderate intellectual disability, and severe behavioral dysregulation: A case report. *SAGE open medical case reports*. 2026. PMID 41625139. DOI 10.1177/2050313X261419571.
6. Meza N, Franco JV, Sguassero Y, et al. Atypical antipsychotics for autism spectrum disorder: a network meta-analysis. *The Cochrane database of systematic reviews*. 2025 May 21. PMID 40396498. DOI 10.1002/14651858.CD014965.pub2.
7. Limbert M, Condessa L, Cardoso C, et al. Hyperthermia in a Teenager: Is it Just a Heat Stroke? *Acta medica portuguesa*. 2026 Aug 3. PMID 42361318. DOI 10.20344/amp.24465.
8. Abhilasha P, Sharma M. Recent Advances in Pharmacological Management of Autism: A Narrative Review. *Cureus*. 2026 Jan. PMID 41664794. DOI 10.7759/cureus.101053.
9. Brown KR, Helvey CI, Kranak MP, et al. Functional Analysis Decision-Making Considerations. *Behavior analysis in practice*. 2025 Dec. PMID 41523829. DOI 10.1007/s40617-025-01057-w.
10. Ugwuocha NW, Neilson S, Challinor A, et al. De-Escalation in United Kingdom Inpatient Mental Health Care: A Structured Critical Narrative Synthesis of People, Capability and Context. *International journal of mental health nursing*. 2026-10-01. PMID 42775399. DOI 10.1111/inm.70364.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- dose page for atomoxetine
- dose page for clonidine
- dose page for aripiprazole
- FDA drug label: Risperidone (label effective 2026-06-25)

Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.

Figure credits

FIGURE 1: High-Dose Propranolol for Severe and Chronic Aggression in Autism Spectrum Disorder: A Pilot, Double-Blind, Placebo-Controlled, Randomized Crossover Study. Randomized Controlled Trial. PMID 39174017, DOI 10.1097/JCP.0000000000001895, PubMed Central (PMC11460740). Cited in this report as [3]. Licence: CC BY-NC-ND 4.0; original image, unaltered.

Each figure shows one study's result; it does not by itself establish that a treatment works or justify a recommendation.